Does the Bond Market Need a Bank Account?

Forward Measure Consistency and Interest Rate Pricing

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Abstract

The standard no-arbitrage framework for bond markets assumes the existence of a tradable bank account $B(t)$ as universal numéraire. But B is not directly traded — it must be synthesized from zero-coupon bonds via a rolling strategy that, in continuous time, requires rebalancing through uncountably many maturities. Meanwhile, practitioners price and hedge using T-forward measures Q^T, each derived from a single tradable bond, without invoking B at all.

This raises a question: does the family $\{Q^T\}$ of local pricing measures, built from tradable bonds alone, necessarily assemble into a single global measure Q — or is Q (and therefore B) an additional assumption?

We formalize this question and show it has substance. In discrete time, consistency is automatic: the bank account emerges as a self-financing rolling strategy, and we prove a four-way FTAP equivalence as proof of concept. In continuous time, we provide a precise, Q-free definition of forward-measure consistency (Definition 3.2.1) and state the question in a falsifiable form. We identify the diagonal singularity of the forward rate surface as a possible mechanism for failure, though establishing whether it actually occurs in bond markets remains open.

To demonstrate that the question is not vacuous, we conduct a descriptive empirical exercise: independent calibration of Gaussian HJM models on different maturity segments of the EUR government bond market. The implied market prices of risk differ systematically across segments, in a divergent pattern that the synthetic controls we examined do not replicate. This does not answer the question — the observed pattern is consistent with either absence of a single Q or the inadequacy of the model class used — but it establishes that the question has empirical content.

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1. Introduction

1.1 The Question

The foundational insight of Black and Scholes (1973) and Merton (1973) is that derivative prices are determined by replication using tradable assets. The risk-neutral measure Q, introduced by Harrison and Kreps (1979), encodes this replication logic mathematically (Bingham and Kiesel, 2004).

In bond markets, the tradable assets are zero-coupon bonds $p(\cdot, T)$. The bank account $B(t) = \exp\left(\int_0^t r(s)\,ds\right)$ — the standard numéraire of the FTAP — is not directly traded. To construct it, one must execute a rolling strategy: hold the just-maturing bond, roll the proceeds into the next. In discrete time, this is a finite, self-financing strategy using traded bonds. In continuous time, it requires continuous rebalancing through an uncountable family of maturities — and the convergence of this strategy is not guaranteed by the well-behavedness of individual bond prices.

Practitioners already work without B: to price a caplet at T, they use the T-bond as numéraire and Q^T as pricing measure. Each operation is faithful to the Black-Scholes-Merton logic — replicate using tradable instruments, extract the price. This paper takes the practitioners' approach seriously and asks:

> *Does the family $\{Q^T\}$ of local pricing measures, derived from tradable bonds, necessarily cohere into a single global measure Q?*

If yes, the bank account is a derived object — it emerges from the consistency of local pricing, and the standard FTAP is confirmed from the bottom up. If no, the standard theory requires an assumption (the existence of Q and B) that goes beyond what tradable instruments can deliver.

In discrete time, we show the answer is yes: consistency is automatic and B is constructible (Section 2). In continuous time, we formalize the question precisely and leave it open (Section 3). We then present a descriptive empirical exercise showing the question has empirical content (Section 4).

To be clear about what is and is not established: we contribute (i) a precise, Q-free definition of forward-measure consistency (Definition 3.2.1), under which the pairwise conditions provably self-propagate (Lemma 2.4.1) and exclude bond-ratio bubbles; (ii) the identification of the numéraire B — not measure incompatibility, and not a projective-limit problem — as the sole locus of the continuous-time obstruction (Section 3.4); and (iii) a descriptive demonstration that the question has empirical content (Section 4). We do not resolve the continuous-time existence question, and we do not claim a relationship to existing large-market results beyond a precise statement of the residual gap (Section 5).

1.2 Related Work

Musiela and Rutkowski (1997) and Döberlein et al. (2000) proved uniqueness of the implied savings account — but took existence as given. Our question is the complement: existence given local data.

The closest existing work is Klein et al. (2016), who investigate bond markets where “the bank account process is not a valid numéraire or does not exist at all.” They argue this is “not the exception but rather the rule” when starting from terminal bonds as numéraires. Using large financial market methods, they prove that no asymptotic free lunch (NAFL) is equivalent to the existence of an equivalent local martingale measure with respect to a terminal bond, and they construct a generalized bank account as a limit of convex combinations of roll-over bonds. Their work establishes the theoretical framework for bond markets without a classical bank account. Our question starts from a different point — pairwise measure consistency (Definition 3.2.1) rather than a no-arbitrage condition on trading strategies — and the relationship between these two starting points is, to our knowledge, not established. Klein et al.’s generalized bank account is a limit object that may differ from the classical B; whether pairwise consistency of $\{Q^T\}$ implies NAFL (or vice versa) is an open question that connects our work to theirs.

Herdegen (2017) develops a numéraire-independent FTAP for general markets. Whether his condition $\mathrm{NA}^{\mathrm{ni}}$ coincides with, is weaker than, or is stronger than our Bond-Family condition (BF) in continuous-time bond markets is, to our knowledge, open.

A continuous-time bond market with a continuum of maturities is a large financial market in the sense of Cuchiero et al. (2016), De Donno (2004), Takaoka and Schweizer (2014), and Carmona and Tehranchi (2006). Our question — does $\{Q^T\}$ cohere into a single Q? — is related to the existence of an equivalent martingale measure in such markets. We note the connection but do not pursue it formally; the bond-specific structure may give the question a different character from the general large-market problem, though we cannot establish this precisely.

Filipović (2001, 2009) studies consistency problems for HJM models — when HJM dynamics preserve the shape of the yield curve within a given parametric family. This is a related but distinct question: Filipović asks whether a model is internally consistent with a family of initial curves, while we ask whether independently calibrated local measures are mutually consistent. The tools developed in Filipović’s framework (function spaces, regularity conditions) may be relevant to resolving our question, but we do not establish the connection formally.

Henrard (2007) and Morini (2009) identified the post-2007 failure of single-curve pricing. The multi-curve literature (Filipović and Trolle, 2013; Crépey et al., 2015; Grbac and Runggaldier, 2015) addresses a different mechanism — counterparty credit risk in interbank rates — but the conceptual parallel is suggestive: the single-curve assumption is the assumption that $\{Q^T\}$ coheres.

In retrospect, our question belongs to what has been called the bottom-up or model-free tradition in mathematical finance: Samuelson (1938) asked whether observed choices imply preferences exist; Hobson (1998) and the model-free finance programme (Davis and Hobson, 2007; Acciaio et al., 2013) ask what can be inferred from observed prices without assuming a model; Balbás et al. (2002) use projective limits of measure families to characterize no-arbitrage over infinite horizons. We recognized this connection during the development of the paper, not at the outset.

1.3 Structure

Section 2 establishes the proof of concept: in discrete time, forward-measure consistency is automatic. Section 3 formalizes the continuous-time question. Section 4 presents a descriptive empirical exercise demonstrating the question has empirical content. Section 5 identifies directions for future investigation.

2. Discrete Time: Proof of Concept

2.1 Setup

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t=0,\dots,N}, \mathbb{P})$ be a filtered probability space. The market consists of default-free zero-coupon bonds $p(t,T)$ with $p(T,T) = 1$ and $p(t,T) > 0$, plus risky assets $X^1, \dots, X^d$. No bank account is assumed.

A trading strategy is a predictable process specifying holdings in each asset. It is self-financing if portfolio value changes arise solely from price movements. An arbitrage is a zero-cost self-financing strategy producing $V_N \geq 0$ a.s. with $\mathbb{P}(V_N > 0) > 0$.

2.2 The Rolling Bond Strategy

Proposition 2.2.1. Define $B(0) := 1$ and $B(t) := \prod_{s=1}^{t} 1/p(s-1,s)$. Then B is the value process of a self-financing strategy holding, during each period $(t-1,t]$, exactly $B(t-1)/p(t-1,t)$ units of the just-maturing bond $p(\cdot,t)$. The one-period return $1/p(t-1,t)$ is deterministic given $\mathcal{F}_{t-1}$.

2.3 The Four Equivalences

Proposition 2.3.1 (Bond-Based FTAP, Discrete Time). The following are equivalent:

(NA) No arbitrage exists in the bond market.

(EMM) $\exists Q \sim P$ such that $S(t)/B(t)$ is a Q-martingale for all traded assets S.

(RF) $\exists Q \sim P$ such that $E^Q[R_t^S \mid \mathcal{F}_{t-1}]$ equals the just-maturing bond return $[1-p(t-1,t)]/p(t-1,t)$ for all traded S, where $R_t^S := S(t)/S(t-1)-1$ is the one-period simple return.

(BF) For each $T \in \{1,\dots,N\}$, $\exists Q^T \sim P$ such that $S(t)/p(t,T)$ is a Q^T-martingale for all traded assets S and $t \le T$.

Moreover, the forward measures constructed in the proof of (EMM) $\Rightarrow$ (BF) satisfy the consistency relation

$$\left.\frac{dQ^{T_1}}{dQ^{T_2}}\right|_{\mathcal{F}_{T_1}} = \frac{p(0,T_2)}{p(0,T_1)\cdot p(T_1,T_2)} \qquad (2.1)$$

involving only bond prices, with no reference to B.

Proof sketch. (NA) $\iff$ (EMM): Rolling construction makes B tradable; Harrison and Pliska (1981) applies. (EMM) $\iff$ (RF): Algebraic reformulation — the equivalence follows from multiplying both sides of the martingale condition by $B(t-1)/S(t-1)$ and using $B(t)/B(t-1) = 1/p(t-1,t)$. (EMM) $\Rightarrow$ (BF): Bayes' formula with density $Z_t^T := p(t,T)/[B(t)\cdot p(0,T)]$. (BF) $\Rightarrow$ (NA): Under Q^N, $V_t^\varphi/p(t,N)$ is a martingale; $V_0 = 0$ forces $V_N = 0$ a.s. Full proofs in Appendix A. $\square$

Remark. The proposition assembles standard ingredients (Harrison and Pliska for (NA) $\iff$ (EMM), Bayes' formula for the numéraire change, Geman et al. 1995 for the general framework). Its purpose here is not mathematical novelty but to establish the four-way equivalence and to identify the precise point where the argument uses discreteness: the chain rule across maturity triples is a finite number of conditions. This is the point that becomes problematic in continuous time.

2.4 A Consistent Family Always Exists — But Only in Discrete Time

Given (EMM), the family $\{Q^T\}$ constructed via $Z_t^T = p(t,T)/[B(t)\cdot p(0,T)]$ satisfies (2.1), and the chain rule

$$\frac{dQ^{T_1}}{dQ^{T_3}} = \frac{dQ^{T_1}}{dQ^{T_2}} \cdot \frac{dQ^{T_2}}{dQ^{T_3}}$$

across maturity triples is a finite number of conditions, each satisfied by construction. In an incomplete market other forward-measure selections exist; (2.1) singles out the mutually compatible family.

The discreteness that does the work here is discreteness of *trading*: B is a finite product of one-period bond returns. With finitely many maturities but continuous trading, B still need not exist for all t (Section 3.4). That a consistent family glues into a global (Q, B) is, under discrete trading, a consequence of finitude, not of structure.

Lemma 2.4.1 (Pairwise consistency is self-propagating). *Under the true-martingale hypothesis of Definition 3.2.1, if a family $\{Q^T\}$ satisfies (3.1) for every pair, it satisfies the triple relation automatically: for $T_1 < T_2 < T_3$,*

$$\left.\frac{dQ^{T_1}}{dQ^{T_3}}\right|_{\mathcal{F}_{T_1}} = \left.\frac{dQ^{T_1}}{dQ^{T_2}}\right|_{\mathcal{F}_{T_1}} \cdot \left.\frac{dQ^{T_2}}{dQ^{T_3}}\right|_{\mathcal{F}_{T_1}}.$$

Proof. By the true-martingale hypothesis, the bond ratio $p(\cdot, T_2)/p(\cdot, T_3)$ is a true Q^{T_3}-martingale on $[0, T_2]$, so optional sampling at $t = T_1$ is valid and (3.1) for the pair (T_2, T_3) gives $\frac{dQ^{T_2}}{dQ^{T_3}}\big|_{\mathcal{F}_{T_1}} = \frac{p(0,T_3)}{p(0,T_2)} \cdot \frac{p(T_1,T_2)}{p(T_1,T_3)}$. Multiplying by (3.1) for (T_1, T_2) and cancelling $p(T_1, T_2)$ and $p(0, T_2)$ returns (3.1) for (T_1, T_3). $\square$

The cocycle is therefore not an additional requirement: pairwise consistency is never the source of failure. Whatever obstructs the passage to a global Q in continuous time, it is not incompatibility of the measure family at any finite level.

Remark (Relationship to Herdegen 2017). In bond markets with strictly positive prices, Herdegen's numéraire-independent no-arbitrage ($\mathrm{NA}^{\mathrm{ni}}$) coincides with NA in discrete time: with finitely many assets all having strictly positive prices, any numéraire is equivalent to any other, and the no-arbitrage conditions coincide (Herdegen 2017, Proposition 3.5). Our Bond-Family formulation (BF) is the only condition among the four equivalences that extends to continuous time without assuming B. Whether (BF) in continuous time is equivalent to, weaker than, or stronger than $\mathrm{NA}^{\mathrm{ni}}$ restricted to bond markets is an open question.

3. Continuous Time: The Question

3.1 Where the Proof Breaks

In continuous time, the rolling strategy requires trading at every instant through uncountably many bonds (cf. Björk et al. 1997, Döberlein and Schweizer 2001). The Bond-Family condition survives — each Q^T uses only the tradable T-bond — but consistency over a continuum of maturities is no longer automatic.

3.2 Forward-Measure Consistency

We define the key concept without presupposing Q or B.

Definition 3.2.1 (Forward-Measure Consistency). Let $\{Q^T\}_{T\in\mathcal{T}}$ be a family of probability measures on $(\Omega, \mathcal{F})$, each equivalent to $\mathbb{P}$, indexed by a set of maturities $\mathcal{T} \subseteq [0, T^*]$. Assume that for each T, $S(t)/p(t, T)$ is a Q^T-local martingale on $[0,T]$ for all traded assets S, including bonds of all maturities (the forward martingale property), and that for each pair $T_1 < T_2$ the bond ratio $p(\cdot,T_1)/p(\cdot,T_2)$ is a *true* Q^{T_2}-martingale on $[0,T_1]$ — equivalently, no bond-ratio bubble arises up to the shorter maturity. The family is *consistent* if for all $T_1 < T_2$ in $\mathcal{T}$:

$$\left.\frac{dQ^{T_1}}{dQ^{T_2}}\right|_{\mathcal{F}_{T_1}} = \frac{p(0,T_2)}{p(0,T_1)\cdot p(T_1,T_2)}. \quad (3.1)$$

A remark on the status of (3.1). It is a genuine condition, not a consequence of the bare forward martingale property. Under Q^{T_2}, the change of numéraire from the T_2-bond to the T_1-bond produces *a* measure with density (3.1) — a valid T_1-forward measure. But in an incomplete market forward measures are not unique, and the family member Q^{T_1} need not coincide with this numéraire-change image. Consistency is precisely the requirement that the family's selections are mutually compatible. Note that the true-martingale clause in the definition's standing assumption is exactly what makes the right-hand side of (3.1) a bona fide Radon-Nikodym density: for the pair (T_1,T_2) the bond ratio $p(\cdot,T_1)/p(\cdot,T_2)$ is a non-negative Q^{T_2}-local martingale with terminal value $1/p(T_1,T_2)$, hence a supermartingale, and its expectation equals $p(0,T_1)/p(0,T_2)$ — so (3.1) integrates to one — if and only if it is a true martingale rather than a strict local one. A strict local martingale (a bond-ratio bubble) would make the right-hand side integrate to less than one. The definition therefore *presupposes* the absence of bond-ratio bubbles up to each shorter maturity; it does not derive it. (On strict-local- martingale bubbles and their characterisation via change of numéraire, see Cox and Hobson 2005, Jarrow, Protter, and Shimbo 2010, and Protter 2013.)

Remark. When $\mathcal{T}$ is finite, condition (3.1) suffices: Proposition 2.3.1 constructs Q and B from the family. By Lemma 2.4.1 the pairwise conditions are mutually compatible at every finite level even when $\mathcal{T}$ is a continuum — so the obstruction to a global Q, if any, is *not* incompatibility of the measure family, nor a projective-limit extension problem (every Q^T already lives on the same $(\Omega,\mathcal{F})$; nothing must be extended to a limit space). What a global risk-neutral measure requires beyond the family is a *numéraire process* B: a single $Q \sim \mathbb{P}$ and a strictly positive adapted B with $\left.\frac{dQ^T}{dQ}\right|_{\mathcal{F}_t} = \frac{p(t,T)}{B(t)p(0,T)}$ for every T. The risk-neutral measure is not any single Q^T; it is anchored by B. Identifying when such a B can be constructed is, to our understanding, the core of the continuous-time problem.

Why standard extension theorems do not immediately apply. A natural reaction is that the Kolmogorov extension theorem (or its measure-theoretic variants) should resolve the question: given a consistent family of finite-dimensional distributions, the theorem constructs a measure on the projective limit. However, the pairwise consistency condition (3.1) is not of Kolmogorov type. Kolmogorov consistency relates measures on *nested sub-σ-algebras* via restriction: $\mu_{n+1}|_{\mathcal{F}_n} = \mu_n$. Our condition (3.1) relates measures on the *same* σ-algebra $\mathcal{F}$ via Radon-Nikodym derivatives determined by bond prices. These are structurally different: (3.1) specifies how the measures *reweight* each other, not how they *restrict*. The passage from pairwise reweighting to a global measure requires constructing a process B such that each Q^T arises as a change of numéraire from Q — a stronger condition than mere compatibility of restrictions. We do not know whether existing projective limit frameworks (such as Balbás et al. 2002) can be adapted to this setting.

3.3 The Question, Precisely Stated

The question of this paper:

> *Given a family $\{Q^T\}_{T\in[0,T^*]}$ satisfying Definition 3.2.1, does there exist a probability measure $Q \sim \mathbb{P}$ and a strictly positive adapted process B such that $S(t)/B(t)$ is a Q-local martingale for all traded assets, with each Q^T being the T-forward measure induced by Q?*

In discrete time the answer is yes (Section 2). In continuous time it is, to our knowledge, open.

We note that in the standard HJM framework driven by finite-dimensional Brownian motion, the existence of Q is equivalent to the HJM drift condition $\alpha(t,T) = \sigma(t,T) \cdot \int_t^T \sigma(t,s)\,ds$. The "if" direction is textbook (Heath et al. 1992). Whether forward-measure consistency *forces* the drift condition — the "only if" direction — is the non-trivial question. We state this as an open problem, not a hypothesis we are in a position to resolve. The restriction to finite-dimensional driving noise is substantive; under infinite-dimensional noise (cylindrical Brownian motion), the question may take a different form.

3.4 Where the Obstruction Lives: the Numéraire B

By Lemma 2.4.1 the measure family is internally consistent at every finite level, so the continuous-time question reduces to a single object: can the numéraire B be constructed? Two features make this non-automatic, and both are about *continuous trading*, not about the cardinality of the maturity set.

First, $B(t) = \exp\left(\int_0^t r(s)\,ds\right)$ requires the short rate $r(s) = f(s,s)$ at every instant. The discrete-time rolling strategy (Proposition 2.2.1) builds B as a finite product of just-maturing bond returns; in continuous time it becomes a continuously rebalanced strategy whose convergence is not guaranteed by the well-behavedness of individual bonds. This is a continuous-*trading* phenomenon: even with finitely many maturities, B need not be constructible for all t.

Second, $r(s) = f(s,s)$ is the diagonal restriction of the forward surface $f(s,T)$. A pointwise diagonal evaluation requires more regularity than the integrated quantities $p(t,T) = \exp(-\int_t^T f(t,u)\,du)$ that the market actually quotes. We note this as a *heuristic analogy* with the trace problem in functional analysis (where restricting a function to a lower-dimensional set requires Sobolev regularity above a threshold), not as an established mechanism: the regularity in question is that of the *joint* surface near the diagonal $T = t$, and we are aware of no evidence that EUR forward surfaces are irregular in this sense — indeed their evident smoothness in the maturity direction (which is what makes parametric fits such as Svensson viable) cuts the other way. We stress, however, that the relevant regularity is the *joint* behaviour of $(s,T) \mapsto f(s,T)$ as $T \to s$ — the diagonal-approach regularity — which is not the same as marginal smoothness in T at fixed s; smoothness of each maturity slice does not by itself control how the slices behave in the limit $T \to s$. We flag the diagonal as the natural place to look for an obstruction, while making no claim that one exists.

Finally, a logical caveat. Failure to construct B would not, by itself, mean the family $\{Q^T\}$ violates (3.1) — by Lemma 2.4.1 it cannot. It would mean the consistent family does not arise from any single Q via a numéraire change. Whether a consistent family can exist *without* an underlying (Q,B) is precisely the open question; the discrete case shows it cannot happen there, and we do not know whether the continuum admits it.

4. Empirical Exercise

4.1 Purpose and Limitations

The preceding sections establish that forward-measure consistency is automatic in discrete time but open in continuous time. A natural question is whether the issue is purely theoretical or whether it has empirical content — whether real bond market data exhibit patterns that are relevant to the question.

We address this with a descriptive exercise, not a formal test of whether Q exists. The exercise operates within a specific parametric model class (Gaussian HJM with maturity-independent market price of risk) and asks whether independently calibrated models on different maturity segments produce compatible risk-price estimates. Incompatibility is consistent with either (a) absence of a single Q, or (b) inadequacy of the model class — specifically, a single Q with maturity-dependent $\lambda(t,T)$ could produce the same pattern (Duffee, 2002; Joslin et al., 2011). We cannot distinguish these alternatives from a single exercise. What the exercise can establish is that the question is not vacuous: the data exhibit systematic cross-segment patterns that a consistency-based diagnostic can detect.

We stress the distance between the theoretical object and the empirical one. Condition (3.1) is a statement about measures; segment-wise equality of λ is a statement about a *bundled proxy* — it presupposes the Gaussian-HJM class *and* a maturity-independent market price of risk, and failure can originate in any of the three components, only the first of which is our question. The exercise therefore establishes "empirical content" only in the weak sense that a consistency-flavoured diagnostic detects systematic cross-segment structure; it does not, and cannot, isolate (3.1). Moreover the ECB curves are Svensson-fitted spot rates, not raw bond prices; the Svensson form imposes cross-maturity smoothness that interacts with segmentation in an unknown direction and could either attenuate or manufacture the cross-segment λ gaps that are the entire result. We report the gaps as a descriptive finding under these explicit caveats.

4.2 Data and Calibration

We use daily zero-coupon yield curves for EUR-denominated government bonds published by the European Central Bank (ECB) over the period January 2014 to December 2024 (2,805 trading days). The dataset comprises 12 maturities: 3 months, 6 months, and 1, 2, 3, 5, 7, 10, 15, 20, 25, and 30 years. The ECB publishes these as spot rates derived from a fitted Svensson parametrization, not raw bond prices; this introduces a layer of interpolation and smoothing that may affect cross-segment calibration results.

We partition the maturity spectrum into overlapping segments $\mathcal{S}_T = [0, T]$ for $T \in \{5, 10, 15, 20, 25, 30\}$ years. For each segment and each trading day, we calibrate a Gaussian HJM model independently using a two-stage procedure. In the first stage, we extract volatility parameters (σ, κ) from a rolling 60-day PCA of bond log-return covariances: eigenvalues give factor variances, and the exponential decay structure $v_k(\tau) \propto (1 - e^{-\kappa_k \tau})$ is fitted to each eigenvector via constrained optimisation (L-BFGS-B, convergence tolerance 10^{-8}, maximum 500 iterations). In the second stage, given (σ, κ), the market prices of risk λ are estimated by cross-sectional OLS regression of observed yields on the model basis functions $g_k(\tau) = (\sigma_k/\kappa_k)(1 - e^{-\kappa_k \tau})$.

This two-stage procedure is a sequential estimator: estimation error in (σ, κ) propagates into λ (the generated-regressor problem; Pagan 1984). The standard errors reported below account for serial correlation (Newey-West HAC, automatic bandwidth selection, yielding bandwidths of 10–15 lags) but not for first-stage estimation error. The conclusions of this exercise rest on the magnitude and geometry of λ gaps rather than on marginal significance.

If a single Q with maturity-independent λ exists, then λ must be identical regardless of which segment is used for calibration. Systematic discrepancies indicate either that the model class is inadequate or that the assumption of a single maturity-independent λ is wrong — or both.

Two-factor specification. The baseline model uses two volatility factors:

$$\sigma_1(\tau) = \sigma_1 e^{-\kappa_1 \tau}, \qquad \sigma_2(\tau) = \sigma_2 e^{-\kappa_2 \tau}, \quad (4.1)$$

with bounds $\kappa_1 \in [0.01, 2]$, $\kappa_2 \in [0.05, 2]$, $\sigma_i \in [0, 0.05]$, $\lambda_i \in [-2, 2]$.

Three-factor specification. We augment with a curvature factor:

$$\sigma_3(\tau) = \sigma_3\, \tau\, e^{-\kappa_3 \tau}, \quad (4.2)$$

which has a hump-shaped profile peaking at $\tau = 1/\kappa_3$. This three-factor decomposition captures the dominant modes of yield curve variation (Litterman and Scheinkman 1991). Additional bounds: $\sigma_3 \in [0, 0.05]$, $\kappa_3 \in [0.05, 2]$.

For each pair of segments $(\mathcal{S}_{T_1}, \mathcal{S}_{T_2})$ with $T_1 < T_2$, we compute the daily gap

$$\Delta\lambda_k(t) = \lambda_k^{(T_1)}(t) - \lambda_k^{(T_2)}(t). \quad (4.3)$$

4.3 Two-Factor Results

Under the two-factor specification, the mean gap in the level market price of risk between the 5Y and 30Y segments is $\overline{\Delta\lambda_1}(5\text{Y} \to 30\text{Y}) = -0.089$. The gap grows monotonically with segment distance: adjacent pairs show small gaps, while the 5Y–30Y pair shows the largest. Long segments consistently produce lower level risk prices than short segments.

The pattern is invariant to parametrization choices. It persists under alternative bounds for the mean-reversion parameter κ_1 (Section 4.4) and under both the two-factor and three-factor specifications (Section 4.5). This monotonicity indicates a genuine feature of the data, even though the absolute magnitude of λ is model-dependent.

Cross-sectional R^2 (the fraction of yield variance across maturities explained by the model on each date, averaged over all dates) ranges from 0.94 (5Y segment) to 0.84 (30Y segment). The 30Y fit would not typically be flagged as inadequate by standard calibration diagnostics — yet the λ gaps are substantial.

4.4 Robustness: Mean-Reversion Bound Sensitivity

The level factor's mean-reversion parameter κ_1 is estimated at or near its lower bound for segments of 10 years and above on more than 95% of trading days — a known identification issue for near-integrated level factors. To check whether the λ_1 gap is an artefact of this, we re-run with the lower bound raised from 0.01 to 0.05. The results are unchanged: the monotonic pattern and gap magnitudes are preserved.

4.5 Three-Factor Comparison

If the two-factor λ gaps reflect only model inadequacy, a three-factor model should reduce them uniformly. The data show something different.

Long-end pairs ($\geq$ 15Y). The λ_1 gaps shrink to near zero. The curvature factor absorbs the cross-segment discrepancy entirely. R^2 improves substantially (from 0.84 to 0.94 for the 30Y segment). For these pairs, the two-factor gap reflects a missing factor, not a structural feature of the data.

Short-vs-long pairs (5Y vs $\geq$ 20Y). The λ_1 gaps do not shrink — they widen, from approximately -0.09 to -0.25 to -0.41. The curvature factor is

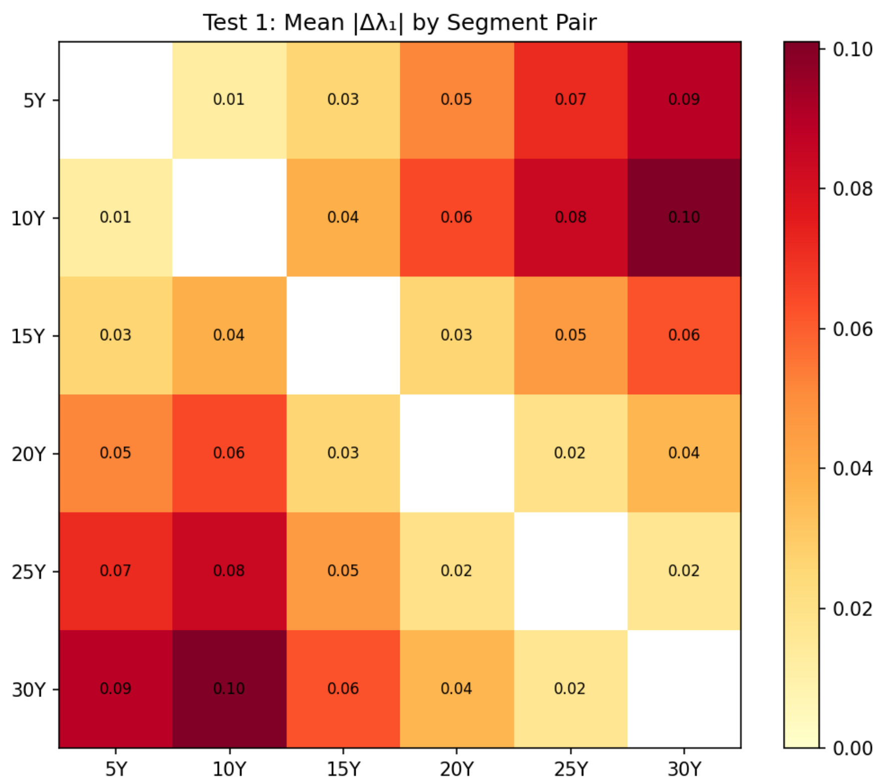

Figure 1: Market Price of Risk Gaps: Heatmap by Segment Pair

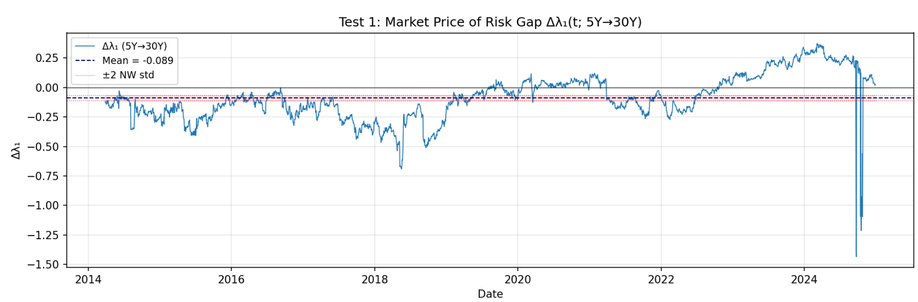

Figure 2: Market Price of Risk Gaps Over Time (5Y vs 30Y)

calibrated with fundamentally different characteristics across these segments ($\kappa_3 \approx 0.47$ for 5Y vs. $\kappa_3 \approx 1.1$–1.6 for 15Y+), suggesting the third factor is doing different work in different parts of the curve.

This divergent pattern — gaps vanishing in one region while amplifying in another — is the most informative feature of the exercise. A uniformly misspecified model predicts uniform gap reduction under enrichment. The data do not show this.

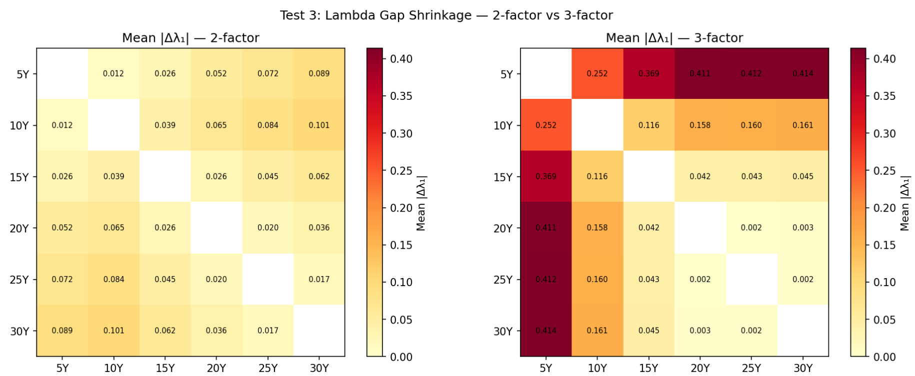

Figure 3: Three-Factor Comparison: Gap Shrinkage by Segment Pair

4.6 Synthetic Control

To assess whether the divergent pattern could arise from the calibration methodology alone — even when a single Q exists by construction — we repeat the exercise on synthetic data.

Design. We generate 2,805 daily yield curves from a three-factor Gaussian HJM with parameters set to the median empirical estimates from the 30Y segment. By construction, a single Q exists and λ is maturity-invariant. We apply the identical segment-wise calibration procedure.

Result. The synthetic three-factor calibration produces λ gaps in the range $|\Delta\lambda_1| \approx 0.01$–$0.09$, uniformly across all segment pairs. This reflects PCA basis heterogeneity — each segment's PCA produces slightly different basis functions, generating apparent gaps even when true λ is constant. The empirical short-vs-long gaps (0.25–0.41) exceed this floor by a factor of 4–20×.

Repeating with parameters from the 5Y segment produces a different synthetic pattern: large gaps with reversed sign ($\Delta\lambda_1 \approx +0.16$ to $+0.22$ for short-vs-long pairs). Neither DGP reproduces the empirical divergent geometry. The synthetic exercise is sensitive to DGP specification, which limits its probative value — but it does establish that the empirical pattern is not a generic artefact of segment-wise calibration.

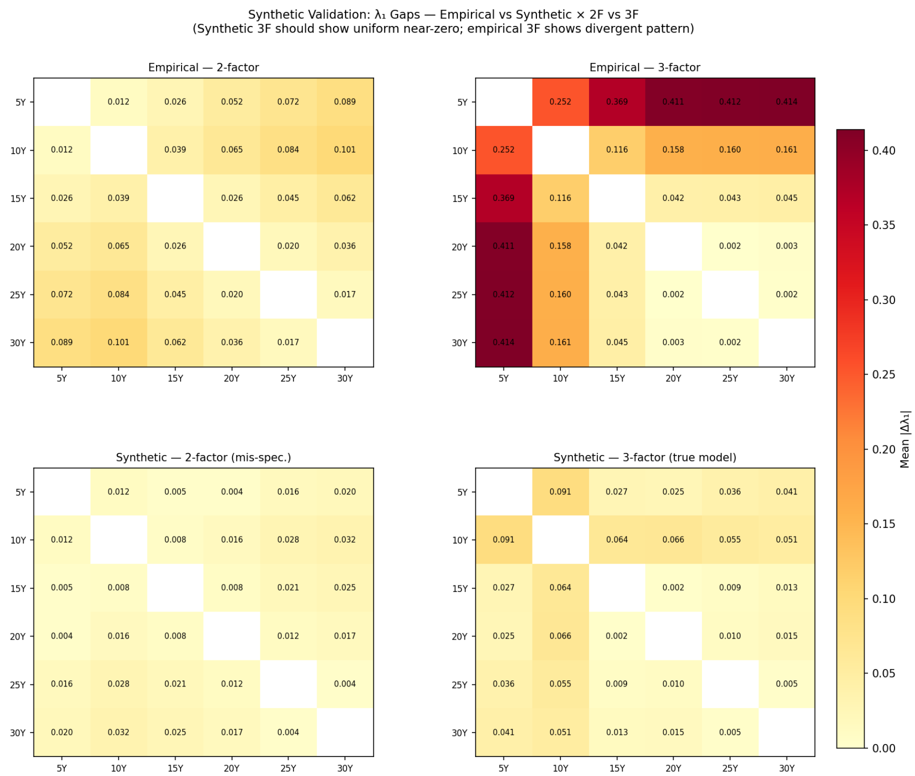

Figure 4: Synthetic vs Empirical: Gap Comparison

4.7 What the Exercise Shows and Does Not Show

The exercise establishes three things:

First, the forward-measure consistency question has empirical content. Within the Gaussian HJM class with maturity-independent λ, cross-segment coherence fails in a systematic, monotonic pattern that standard R^2 diagnostics do not detect.

Second, the three-factor comparison reveals a divergent structure: long-end gaps are absorbed by an additional factor (model inadequacy), while short-vs-long gaps amplify (something the model class cannot reconcile). This divergence is not reproduced by synthetic controls.

Third, the exercise does *not* establish that a single Q fails to exist. A single Q with maturity-dependent $\lambda(t, T)$ — which is precisely what preferred-habitat models produce (Vayanos and Vila 2021) — could generate the observed pattern when projected onto a maturity-independent specification. Distinguishing this from genuine absence of Q would require testing within a model class that permits maturity-dependent λ, which we leave to future work.

In particular, because the synthetic control was examined for only two data-generating parametrisations (30Y- and 5Y-calibrated), and is sign-sensitive across them, the exercise establishes at most that the empirical pattern is not a *generic* artefact of segment-wise calibration — never that it is irreproducible by *some* single-Q model. This ceiling applies to every use of the synthetic evidence in the paper.

The exercise is a proof of concept for the question, not evidence for the answer.

5. Directions

The question formalized in Section 3.3 — whether pairwise forward-measure consistency implies the existence of a global Q — is, to our knowledge, open. We identify three directions for investigation, each requiring expertise beyond the scope of this paper.

The mathematical question. By Lemma 2.4.1 the measure family is consistent at every finite level; the open content is whether a numéraire process B — equivalently a global Q — can be constructed from a consistent family in continuous time (Section 3.4). An explicit counterexample — a consistent family $\{Q^T\}$ in the sense of (3.1) admitting no underlying (Q, B) — would settle the question negatively and would be the most valuable single contribution. In finite dimensions with deterministic volatility the problem may be tractable by functional-analytic methods, with the diagonal $r(s) = f(s, s)$ the natural place to seek an obstruction. The relationship to the no-asymptotic-free-lunch (NAFL) condition of Klein, Schmidt, and Teichmann (2016) is the most promising connection and is discussed separately below.

Richer model classes and sharper diagnostics. The empirical exercise operates within Gaussian HJM with maturity-independent λ. Testing within model classes that permit maturity-dependent λ (Duffee, 2002; Joslin et al., 2011) — or within non-Gaussian specifications (stochastic volatility, regime-switching) — would clarify whether the observed divergent pattern survives or is absorbed by a richer single-Q specification. Two refinements would also sharpen the present diagnostic without changing the model class: bounding the first-stage (PCA) estimation error — for instance by block-bootstrapping the estimation window — so that the synthetic-floor margin can be read against first-stage noise; and partially pinning the direction of the Svensson-smoothing interaction with segmentation by re-fitting on raw-maturity subsets or perturbing the input curves. We report the present exercise under explicit caveats (Section 4) rather than as a calibrated test, and leave these refinements to future work.

The relationship to Klein–Schmidt–Teichmann, and why we do not claim it. Klein et al. (2016) construct a generalized bank account as a limit of convex combinations of roll-over bonds; by Section 3.4 our question *is* a numéraire- construction question, so the two programmes are close, and it is natural to ask whether a consistent family in the sense of (3.1) implies their no-asymptotic-free-lunch (NAFL) condition (after which their construction would supply a generalized B). We do not claim this implication, and we state the obstruction precisely. NAFL is a condition on *trading strategies* in the large financial market generated by all bonds: it controls sequences of admissible portfolios under a topology on terminal wealths. Condition (3.1) is a condition on *measures*: it pins each pairwise density to the change-of-numéraire form and, by Lemma 2.4.1, glues consistently at every finite level. A consistent family yields, for each terminal maturity T^*, a single measure Q^{T^*} under which all bonds are local martingales after deflation by $p(\cdot, T^*)$. This resembles KST's terminal-bond ELMM in form, but differs in scope: KST's ELMM is required to control the full large-market admissibility closure (sequences of admissible strategies and their terminal-wealth limits), whereas the family-derived Q^{T^*} controls only the finite-dimensional reweightings that (3.1) encodes and need not regulate that closure. Condition (3.1) thus says nothing directly about the admissibility class or the asymptotic closure that NAFL governs. The residual gap is thus exactly the passage from finite-level measure consistency to a large-market no-arbitrage condition on strategies; bridging it, to our understanding, amounts to constructing the numéraire, and we leave it open. We flag KST as the most likely route to a resolution, not as an established one.

Separately, the relationship between (BF) and Herdegen's (2017) numéraire-independent no-arbitrage $\mathrm{NA}^{\mathrm{ni}}$ in continuous-time bond markets remains open, as noted in Section 2.4.

6. Conclusion

We asked whether the bond market needs a bank account — whether the family of T-forward measures $\{Q^T\}$, built from tradable bonds, necessarily assembles into a single global risk-neutral measure Q, or whether Q is an additional assumption beyond what tradable instruments can deliver.

In discrete time, the answer is yes: the bank account is constructible as a self-financing rolling strategy, and forward- measure consistency is automatic. In continuous time, the question is open. We formalized it precisely (Definition 3.2.1), identified a possible mechanism for failure (the diagonal singularity), and noted connections to the HJM drift condition, the projective limit framework, and the large-financial-markets FTAP literature.

To demonstrate that the question is not vacuous, we conducted a descriptive empirical exercise. Within the Gaussian HJM class with maturity-independent market price of risk, cross-segment coherence fails in a systematic, divergent pattern: long-end discrepancies are absorbed by an additional curvature factor, while short-vs-long discrepancies amplify. Synthetic controls indicate this pattern is not a generic artefact of the calibration methodology. The exercise does not answer the question — the observed pattern is consistent with either absence of a single Q or inadequacy of the restricted model class — but it establishes that the question has empirical teeth.

If this paper contributes anything, it is the question itself. We hope it may inspire investigation by researchers with the mathematical tools to resolve it.

Acknowledgments

This paper grew out of a puzzle I first encountered during my doctoral work (Universität Ulm, 2008): why does the standard no-arbitrage theory rely on a bank account that is not actually traded? The question lay dormant for eighteen years until the tools existed to revisit it.

The present paper was developed in extended collaboration with Claude (Anthropic, Opus). Claude's contributions included: identifying the original dissertation result as an algebraic identity rather than an independent FTAP, which redirected the investigation toward the genuinely open continuous-time question; surveying and positioning the work relative to the forward-measure, large-financial-markets, and numéraire-independent FTAP literatures; iteratively clarifying the question from an intuitive discomfort into the precise formulation of Definition 3.2.1; designing the segment-wise calibration approach, the three-factor comparison, and the synthetic validation; and the scope decision to present this as a question paper rather than a theorem paper — matching the contribution to what the evidence supports. The editorial work, including structuring, drafting, and responding to reviewer critiques, was conducted jointly throughout.

Data, code, and reproducible results are available at https://github.com/shaohui1977/bond-family-na. All errors remain the author's responsibility.

Appendix A: Proofs for Section 2

We prove Proposition 2.3.1. The argument uses three ingredients: the rolling bond construction makes B tradable; the classical FTAP (Harrison and Pliska 1981) applies; and the numéraire change formula (Geman et al. 1995) connects the four conditions.

Lemma A.1 (Strategy space equivalence). The set of terminal payoffs attainable by self-financing strategies in $\mathcal{S}$ coincides with the set attainable in $\mathcal{S} \cup \{B\}$.

Proof. Any strategy holding B can be replicated by executing the rolling bond strategy of Proposition 2.2.1, which is self-financing in $\mathcal{S}$. □

Lemma A.2 (Numéraire invariance). Let N be a tradable asset with $N(t) > 0$, and let (φ, ψ) be self-financing with $V_0 = 0$. If $S^j(t)/N(t)$ is a Q-martingale for all traded assets S^j, then $V_t/N(t)$ is a Q-martingale.

Proof. The self-financing condition gives $V_t/N(t) - V_{t-1}/N(t-1) = \sum_j \theta_t^j \Delta(S^j/N)_t$, where θ_t^j is predictable and each $\Delta(S^j/N)_t$ is a Q-martingale difference. The sum is therefore a Q-martingale difference. □

Proof of Proposition 2.3.1.

(NA) $\iff$ (EMM): By Proposition 2.2.1, B is the value of a self-financing strategy with $B(t) > 0$. By Lemma A.1, including B does not enlarge the attainable set. The classical FTAP (Harrison and Pliska 1981) gives: NA $\iff$ $\exists Q \sim P$ such that S/B is a Q-martingale for all $S \in \mathcal{S}$.

(EMM) $\iff$ (RF): For any S with $S(t-1) > 0$:

$$E^Q[S(t)/B(t) \mid \mathcal{F}_{t-1}] = S(t-1)/B(t-1)$$
$$\iff E^Q[1 + R_t^S \mid \mathcal{F}_{t-1}] = 1/p(t-1,t) = 1 + R_t^{(\mathrm{rf})}$$
$$\iff E^Q[R_t^S \mid \mathcal{F}_{t-1}] = R_t^{(\mathrm{rf})}.$$

All steps are multiplication by positive $\mathcal{F}_{t-1}$-measurable quantities.

(EMM) $\Rightarrow$ (BF): For each T, define

$$Z_t^T := \frac{p(t,T)}{B(t) \cdot p(0,T)}, \quad \left.\frac{dQ^T}{dQ}\right|_{\mathcal{F}_t} = Z_t^T.$$

Since $p(\cdot, T)/B$ is a Q-martingale, Z^T is a strictly positive Q-martingale with $Z_0^T = 1$, so $Q^T \sim P$. By the abstract Bayes formula, for $s \le t \le T$:

$$E^{Q^T}[S(t)/p(t,T) \mid \mathcal{F}_s] = \frac{E^Q[Z_t^T \cdot S(t)/p(t,T) \mid \mathcal{F}_s]}{Z_s^T} = \frac{E^Q[S(t)/B(t) \mid \mathcal{F}_s] \cdot B(s)}{p(s,T)} = \frac{S(s)}{p(s,T)}.$$

(BF) $\Rightarrow$ (NA): Suppose $V_0 = 0$, $V_N \geq 0$ a.s., $P(V_N > 0) > 0$. By (BF) with $T = N$, $\exists Q^N \sim P$ such that $S/p(\cdot, N)$ is a Q^N-martingale. By Lemma A.2, $V_t/p(t,N)$ is a Q^N-martingale. Then $0 = V_0/p(0,N) = E^{Q^N}[V_N/p(N,N)] = E^{Q^N}[V_N]$. With $V_N \geq 0$ Q^N-a.s., this gives $V_N = 0$ Q^N-a.s., hence P-a.s. Contradiction. $\square$

Consistency relation (2.1). Under (EMM):

$$\left.\frac{dQ^{T_1}}{dQ^{T_2}}\right|_{\mathcal{F}_{T_1}} = \frac{Z^{T_1}_{T_1}}{Z^{T_2}_{T_1}} = \frac{p(T_1,T_1)\cdot p(0,T_2)}{p(0,T_1)\cdot p(T_1,T_2)} = \frac{p(0,T_2)}{p(0,T_1)\cdot p(T_1,T_2)}.$$

Note that $B(T_1)$ cancels. The relation involves only bond prices. $\square$

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